Pallavi Kumari

I am motivated, enthusiastic and precision-focused Aeronautical Engineering with strong academic and technical background in aircraft propulsion, UAVs, aircraft design, aerodynamics, aircraft performance and structural analysis. Proficient in using industry-standard tools such as MATLAB, Solid Works, Solid Edge and CATIA for modelling, simulation and performance optimization. Demonstrates a deep understanding of aerospace systems, with a proven ability to solve complex engineering challenges through analytical thinking and innovation. Committed to advancing aerospace technology through collaborative teamwork, rigorous testing, and continuous improvement in both design and safety.

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EDUCATION

Bachelor of Engineering

East West College of Engineering

2022-2026, *Bangalore, Karnataka*

Courses

-Aeronautical Engineering (present 8th Sem)

- CGPA - 8.5

Intermediate

REVA Independent P.U. College

2020 - 2022, *Bangalore, Karnataka*

Courses

- Karnataka Board
- 76.16%

Matriculation

Army Public School, PRTC

2019 - 2020, *Bangalore, Karnataka*

Courses

- CBSE
- 70.4%

PROJECTS

SHAPING THE FUTURE OF ROCKETRY: INSIGHTS AEROSPIKE NOZZLE PERFORMANCE

- Led a team focused on enhancing aerospike nozzle efficiency through systematic analysis of varying truncation ratios.
- Utilized ANSYS Fluent to conduct computational fluid dynamics (CFD) simulations, evaluating the impact of truncation on key performance parameters such as thrust and flow separation.
- Provided critical insights into optimal design configurations based on simulation results.

DEVELOPMENT OF UNMANNED UNDERWATER VEHICLE

- Developed a biomimetic underwater robot inspired by the octopus arm, focusing on achieving flexible, multi-directional mechanical motion for enhanced underwater maneuverability.
- Created detailed 3D models, assemblies, and motion studies solid edge, ensuring proper kinematics, joint flexibility, and structural feasibility.
- Optimized the design the design for hydrodynamic performance, loading handling and manufacturability, using simulation driven improvement.

INTERNSHIP

TATA Advanced Systems Limited

Concept Design

4 months

TRAINING & CERTIFICATION

Software Training

STK

2024,

A powerful modelling and simulation software used in aerospace, defense engineering. STK allows users to model complex systems like satellites, missiles, aircraft, and more.

BOOTCAMP

2025,

The Academy Innova World (AIW) Bootcamp is a structured, short-duration technical and professional development program *designed to enhance industry readiness among engineering and technology graduates.*

It focuses on both ***domain-specific technical competencies*** *and* ***professional skill augmentation*** *(HR and communication modules).*

CO-CURRICULAR ENGAGEMENT

Certificate of participating in Para Gliding

EWCE, Bangalore

Certificate of Participating in Throwball

EWCE, Bangalore

Certificate of Participating in Aircraft Sketching

EWCE, Bangalore

Certificate of Participating in balsawood aircraft modelling

EWCE, Bangalore

SKILLS

CATIA | ANSYS | AGI/STK | Solid Edge | Solid Works | Open Rocket | MATLAB

LANGUAGES

English

Full Professional Proficiency

Hindi

Native or Bilingual Proficiency

INTERPERSONAL SKILLS

INTEGRITY | TEAM PLAYER | ADAPTABILITY | TIME MANAGEMENT